

Machine Learning - Predicting Procrastination based from Screen Time, Sleep Patterns and Lifestyle Factors

Group 13

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1. COMPARING BOTH MODELS USED

Linear Regression (LR) and Random Forest Regressor (RF) were the two models chosen for this regression project. Both models attempt to predict procrastination scores using screen time, sleep patterns and lifestyle factors, but they use different approaches.

Linear Regression
LR predicts a numerical value using a weighted combination of the input features. It was implemented from scratch using NumPy, without using a model ready to use. The implementation calculates predictions, errors and gradients and then updates the coefficients through gradient descent.

- **Strengths:** simple to understand and provides a useful baseline for regression.
- **Weaknesses:** it assumes a linear relationship between the input features and the target, which limits its ability to represent more complex patterns.

Random Forest Regressor
RF combines multiple decision trees and averages their predictions. The model was implemented using scikit-learn.

- **Strengths:** can capture non-linear relationships and interactions between features. Does not require numerical feature scaling.
- **Weaknesses:** it is more complex to interpret than LR. Its performance depends on the selected parameters.

The models were compared using five-fold-cross-validation on the training data. Two learning rates were evaluated for LR (0,001 and 0,01), while RF was evaluated using 50 and 100 trees. The configuration with the lowest mean validation RMSE was selected for the final evaluation.

The preprocessing steps were performed separately within each validation fold, ensuring that information from the validation data was not used to fit the preprocessing transformations.

CROSS-VALIDATION RESULTS

	MAE	MSE	RMSE	R2
Linear Regression (rate=0.001)	0.1536	0.0505	0.2202	0.9346
Linear Regression (rate=0.01)	0.1037	0.0209	0.1437	0.9730
Random Forest (trees=50)	0.0588	0.0133	0.1142	0.9825
Random Forest (trees=100)	0.0572	0.0128	0.1116	0.9833

The configuration with the lowest mean validation RMSE was selected **(Random Forest)**. This comparison was performed before using the test set, preventing the test result from influencing model selection.

2. FINAL EVALUATION ON THE TEST SET

After selecting the configuration through cross-validation, the chosen model **(Random Forest)** was trained using the complete training set and evaluated on the reserved test set.

The following metrics were calculated:

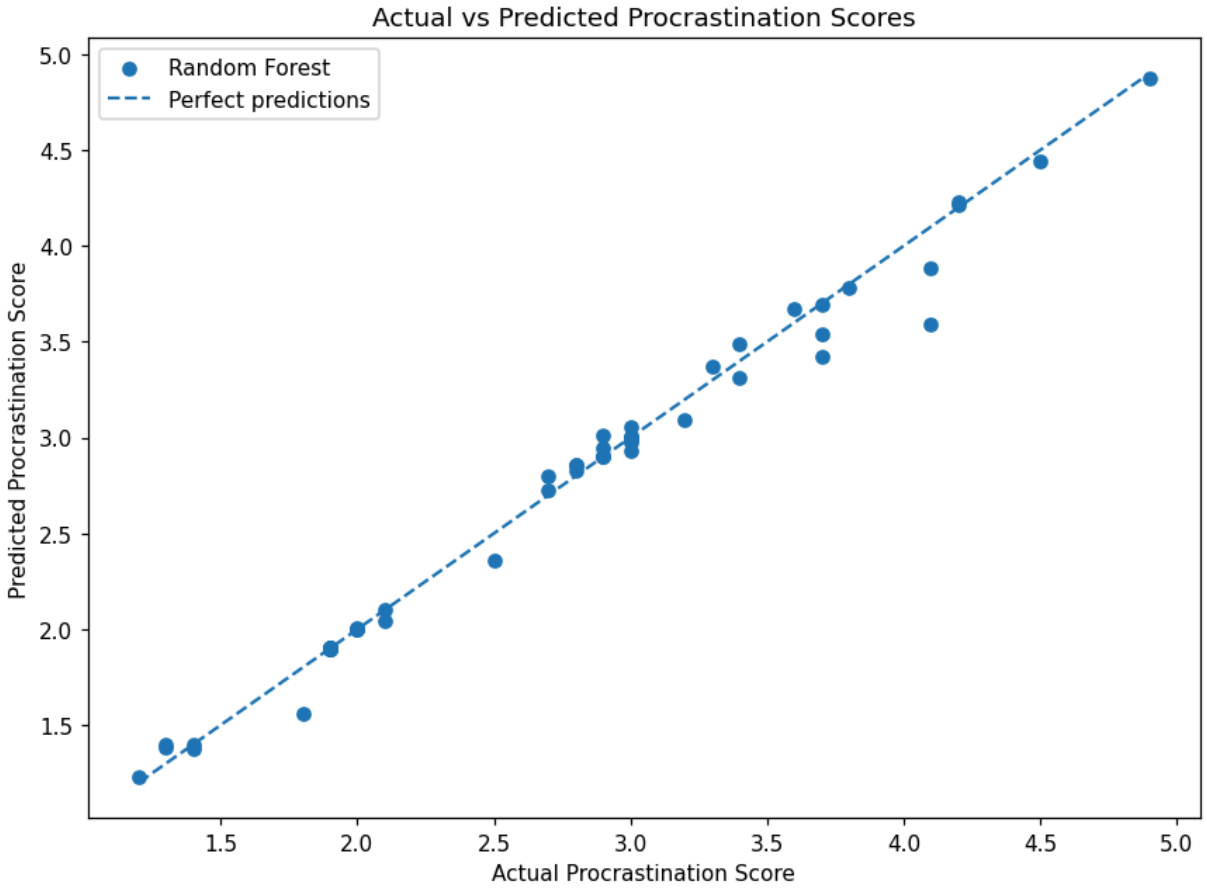
- **Mean Absolute Error (MAE):** measures the average absolute difference between predicted and actual scores. Lower values indicate smaller errors.

- **Mean Squared Error (MSE):** measures the average squared prediction error. Larger errors receive greater weight.

- **Root Mean Squared Error (RMSE):** the square root of MSE. Lower values indicate better predictive performance.

- **R^2:** measures performance relative to predicting the mean target value. Higher values generally indicate better performance.

	MAE	MSE	RMSE	R2
Training	0.0231	0.0020	0.0449	0.9974
Test	0.0506	0.0098	0.0992	0.9865



3. DATASET AND PREPROCESSING

The project used procrastination_survey_dataser_csv, containing 300 observations and 68 columns. The target variable was **procrastination_score**, a numerical variable with values between 1,2 - 4,9.

Twenty predictors related to screen time, sleep, digital habits, daily routines and lifestyle factors were selected. Participant identifiers, metadata, questionnaire items and other derived scores were excluded from the predictors to avoid unnecessary noise to the final prediction.

The CSV was inspected for missing values, duplicate records, inconsistent row lengths and unusual values. An extra field found in some rows was checked and corrected before loading the dataset. The exploration also checked whether daily-hour variables contained negative values or values exceeding 24 hours.

The missing values in **scheduling_method** did not require removing observations from the modelling dataset because that variable was not selected as a predictor.

Categorical values were converted into binary columns using the one-hot encoding. Also, numerical features were standardized for LR using statistics calculated from the training data. RF used the encoded features without scaling.

The dataset was split into training and test sets using a 80-20 proportion. Five-fold-cross-validation was then performed exclusively within the training set.

4. CONCLUSION

The comparison of both models and the selection of the specific model with better results were performed using five-fold cross-validation, while the test set was reserved for the final evaluation. This follows the recommended methodology and reduces the risk of selecting a model based on its test results.

The selected configuration was **Random Forest** with 100 trees with a **mean validation RMSE** of **0,1116** and a **final test RMSE** of **0,0992**.